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GPS-Z 3.0: Tutorial/Walkthrough

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Sep 2026



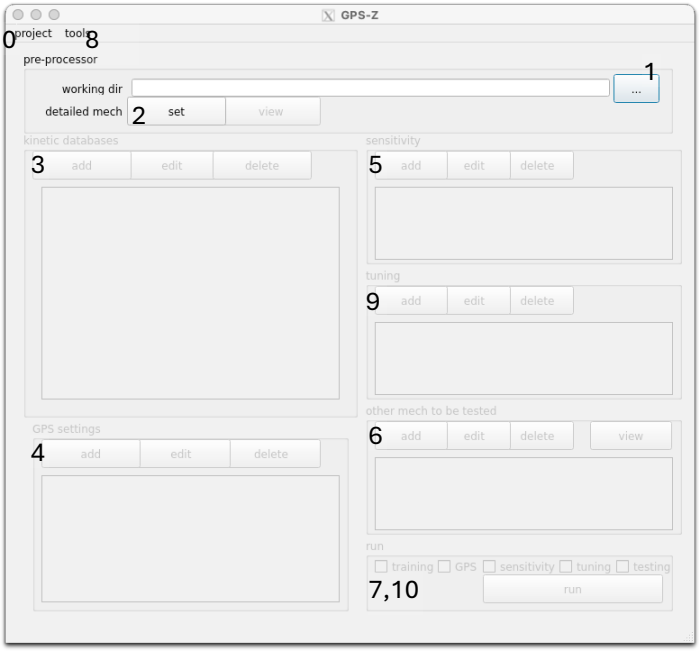
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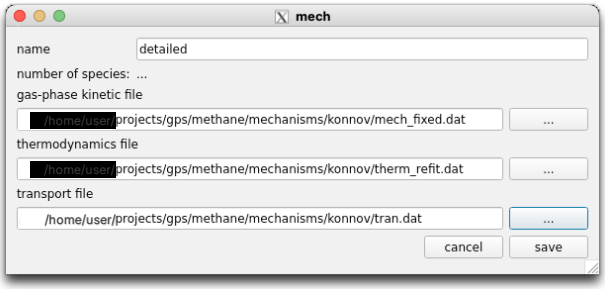
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GPS-Z 3.0: Tutorial/Walkthrough

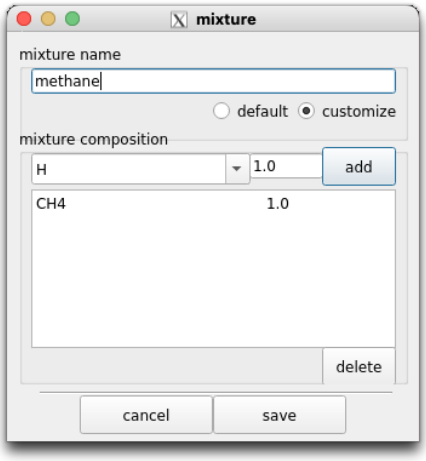
After launching GPS-Z, you should have a window like the one shown here. The number labels on the figure will be referred to in the steps below. To start a new project, click the button next to the working directory text box (1). In the dialog window that opens, select a directory where project files will be stored. The user is advised to save the project whenever making changes using the "project->Save" menu (0). For this tutorial, select the "methane_tutorial" directory included in the GPS-Z repository. This directory contains all the mechanism files that will be used in the tutorial. Mechanism files are copyright their original authors with details included in the respective README files.

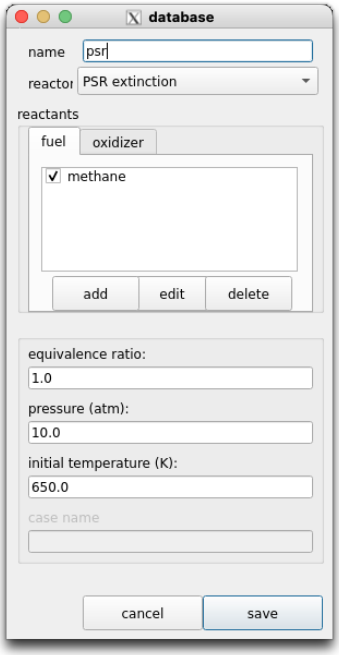


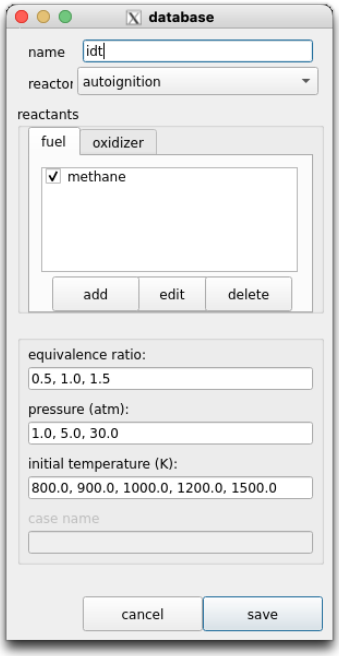


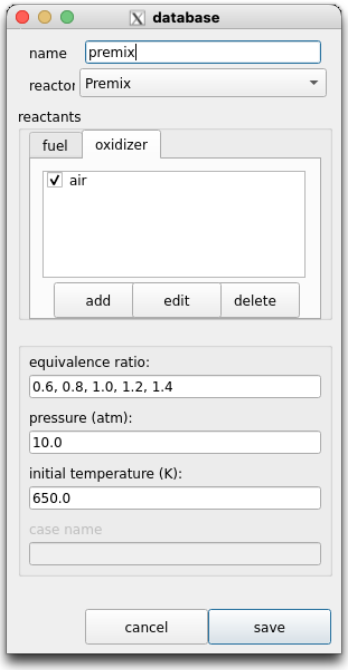
After setting the project directory, we will set the detailed mechanism. Click the "set" button next to "detailed mech" (2) and load the kinetics, thermodynamics, and transport files from the "konnov" directory inside the tutorial directory.

After loading the detailed mechanism, we will create three kinetics databases (3): perfectly stirred reactor (psr), ignition delay time (idt), and premixed flame (premix). These databases will be used for GPS training, reaction rate sensitivity and reaction rate tuning, respectively. Each database will simulate combustion of methane (CH4) with air at selected temperatures, pressures, and equivalence ratios. The tutorial is focused on creating models for accurate prediction of flame speed at 650 K unburned temperature, 10 atm pressure, and a range of equivalence ratios, and the databases are created with relevant conditions. For the first database a fuel must be specified as shown here, and the predefined air mixture can be used for our purposes. Fuel and oxidizer definitions can be reused across databases and multiple fuel/oxidizer combinations can be specified. The images below show the conditions for the three databases we will use.

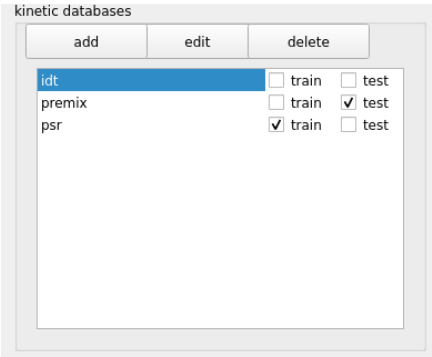


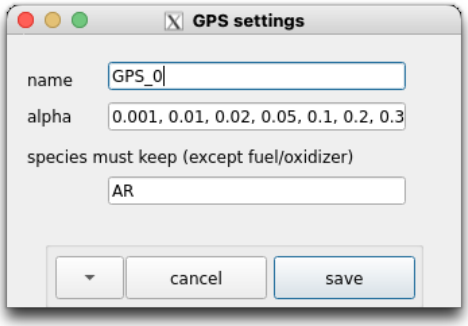






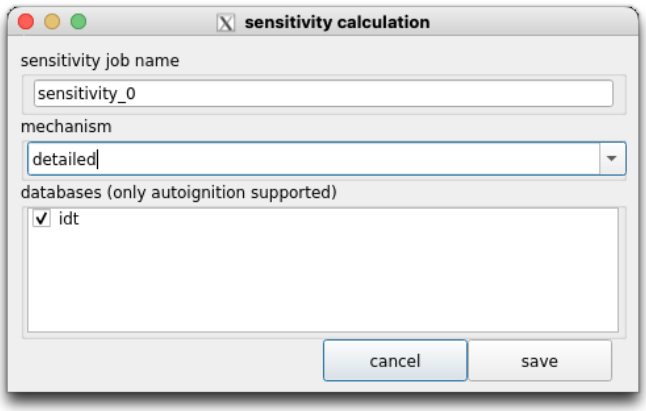
After creating the three databases, select the appropriate check boxes for training and testing as shown here. The “idt” database will be used in the sensitivity analysis later. (Currently sensitivity can only be done with respect to ignition delay time.)



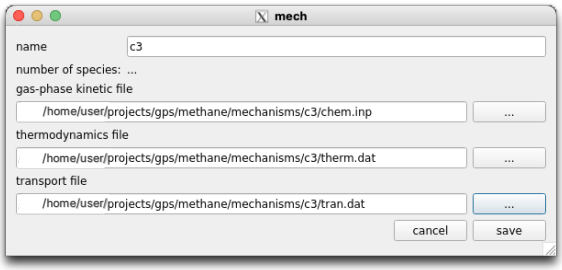


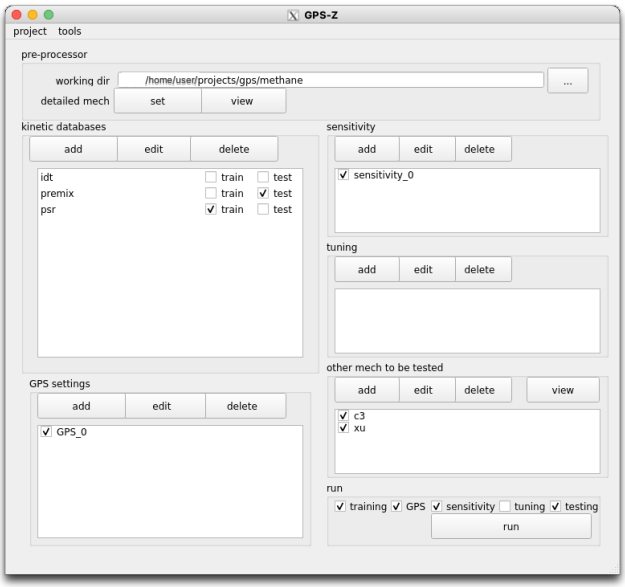
After the databases are set up, we can specify the settings for the GPS algorithm (4). For the tutorial we will use mostly default settings but specify a detailed array of alpha parameters. As seen previously in the database conditions, arrays can be set as comma separated list. The full set of alpha parameters we will use is: “0.001, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95”. This will give us a good range of reduced mechanisms.

Sensitivity calculations are used to ease the selection of reactions for rate tuning. We add a sensitivity job (5), which will use the detailed mechanism and the “idt” database as shown here.

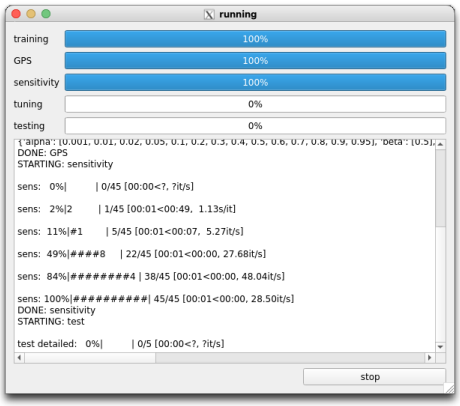


We will compare the results of our reduction and tuning with two recent literature models, using the “other mech to be tested” section (6). We can add in both the “c3” and “xu” models as was done with the detailed model previously.

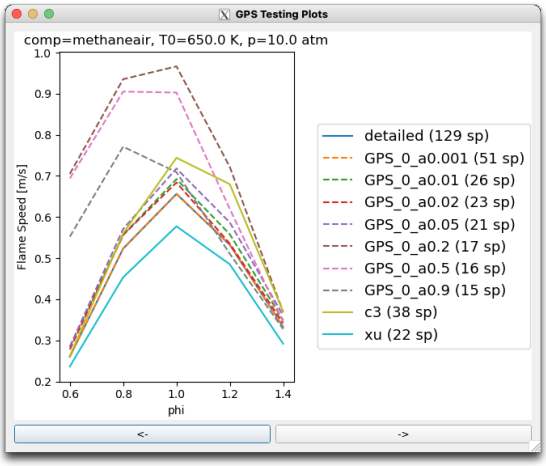


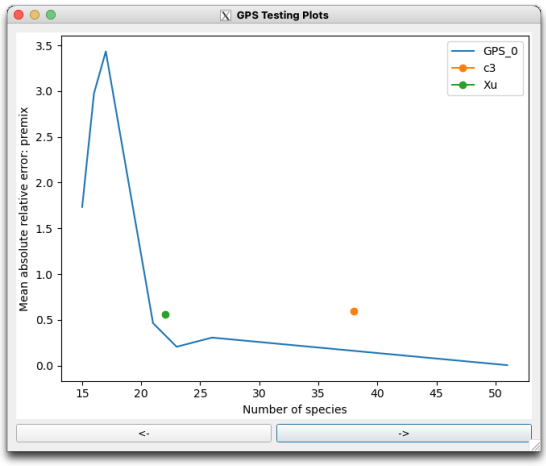


With the databases, GPS settings, sensitivity, and other mech sections completed we can run the “training”, “GPS”, “sensitivity” and “testing” jobs (7). A progress dialog will show the user how the jobs are progressing.

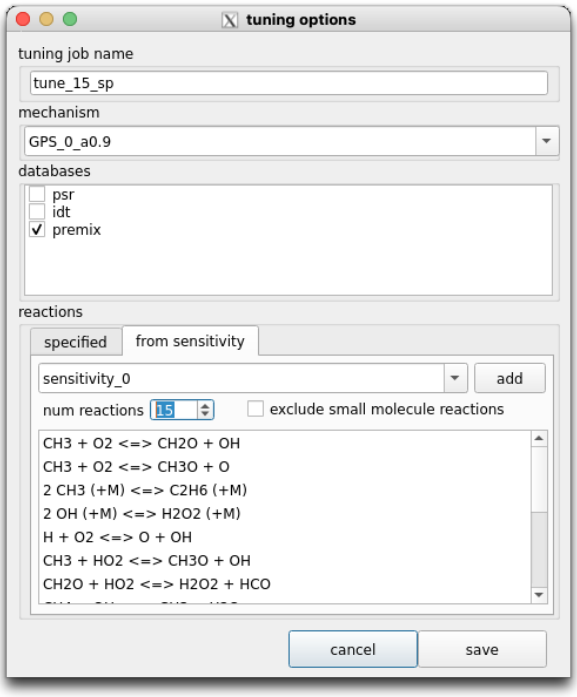


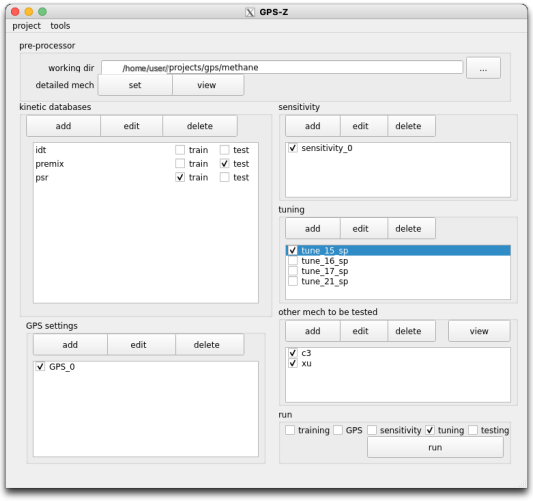
Once the jobs are completed, we can view the testing results using the tools->test plots menu (8). Note that some GPS settings result in identical models and only one model at each size is included in the testing/plotting stage. From the plots below we can see that model performance starts to degrade at around 21 species (alpha 0.05). We will perform tuning on the four smallest models (15, 16, 17, and 21 species) to modify reaction rates to better match flame speed predictions of the detailed model.



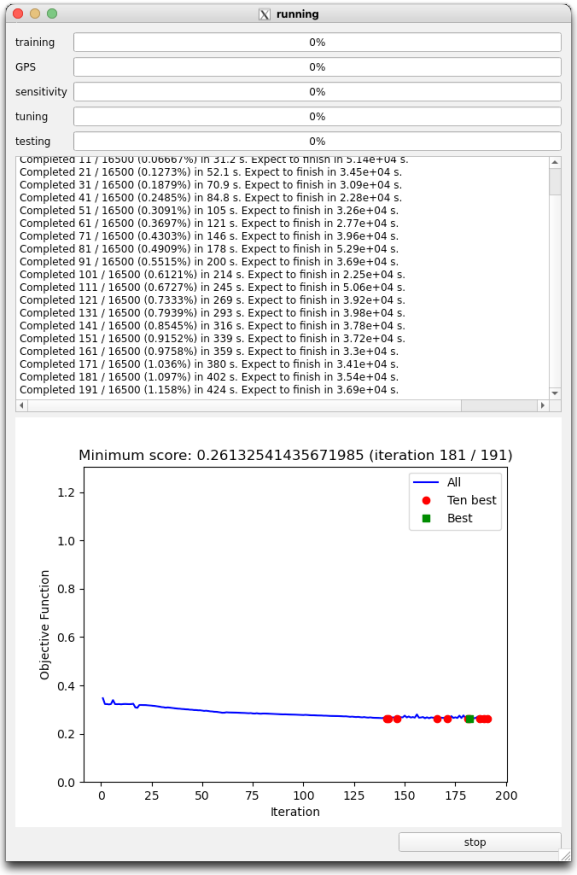


We add four tuning jobs (9) with identical settings except for the mechanism. The example shown here can be used as a guide to set up the tuning jobs. When adding reactions from a sensitivity job, set the number of reactions to use and click the "add" button. After reactions are added from sensitivity, the reactions will be listed in the "specified" tab.





The tuning process takes significantly longer than any other step in this workflow. Each tuning job in this tutorial can be expected to take between 12 and 24 hours to complete. Tuning jobs are only run if they are checked, enabling the user to manage resources and execution time. Additionally, a tuning job can be restarted if interrupted. During job execution for a tuning, the progress window will show a plot of the error objective function.



Finally, once all the tuning jobs are completed, we can re-run the test job (with all tuning jobs checked) and plot the results including the tuned mechanisms. Based on this data, we can select the model that meets our needs with respect to model size and accuracy. The tuned models are saved in the project hierarchy. For example, the GPS_0_a0.05_tuned model is located in: `<project_dir>/GPS/[psr]/K1_b0.5_a0.05/e[default]/tuning/tune_21_sp/mech`

